



Prediction of mechanical properties of carbon fiber/epoxy composite modified by nanoparticles using multiple explainable machine learning algorithms

Akash Baral, Mostofa Aadeeb Chowdhury , Md. Jahid Hasan , Arda Arefin Nir, Manish Bhadra Arnab , Md. Ashraful Islam

Department of Mechanical Engineering, Khulna University of Engineering & Technology, Khulna 9203, Bangladesh

ARTICLE INFO

Keywords:

Epoxy composites
Nanoparticle reinforcement
Machine learning prediction
Explainable artificial intelligence
SHAP analysis

ABSTRACT

The complex interdependencies between constituent materials and processing conditions of Carbon fiber-reinforced epoxy composites enhanced with nanoparticles hinder the accurate prediction of their performance. This study introduces an interpretable machine learning framework to predict tensile and flexural strength in nanomodified CF/epoxy composites. A comprehensive dataset comprising 174 tensile and 108 flexural data points was curated from 50 peer-reviewed studies, encompassing 12 key input parameters such as fiber layers, nanoparticle size and type, and processing conditions. Four regression models were trained using 5-fold cross-validation and assessed via R^2 , MAE, and RMSE. The Decision Tree and Gradient Boosting models achieved the highest test R^2 scores of 0.983 and 0.931 for tensile and flexural strength, respectively, while XGBoost showed consistent performance across both. To overcome the black-box nature of these models, SHapley Additive exPlanations (SHAP) were employed to elucidate feature importance and interactions. For tensile strength, fiber layer count, sonication temperature, and curing temperature were dominant contributors, while flexural strength was most influenced by curing duration, sonication duration, and particle type. This integrative approach not only enhances predictive accuracy but also delivers actionable insights into process-property relationships, paving the way for data-driven composite design and accelerated materials development in advanced engineering applications.

1. Introduction

The innovation of carbon fiber reinforced polymer (CFRP) composites is one of the most significant breakthroughs in material science, adding excellent mechanical properties, lightweight, and high degree of design freedom. Owing to their high strength-to-weight ratio and good fatigue and corrosion properties, these materials have been widely used in aerospace, automotive, marine, and civil engineering, etc. [1,2]. In various fields, including aerospace and defense, the demand for superior mechanical properties, such as fracture toughness, resistance to impact, and fatigue life, has become increasingly important [3]. To overcome these challenges, the use of nanotechnology has become more popular, in which nanoparticles are incorporated in the CFRP composite matrix to obtain synergistic improvements in properties [4].

Recently, researchers have also focused on improving the mechanical and tribological behavior of thermosetting epoxy composites by

incorporating inorganic nanoparticles FeTiO_3 and SiO_2 , using ultrasonic dispersion and compression molding techniques [5]. Carbon nanotubes (CNTs), graphene, nano-silica, titanium dioxide (TiO_2), nanoclays, and other nanomaterials have been studied extensively as reinforcements to improve the performance of the CFRP composites [6,7]. These particles can interact with the polymer matrix at the atomic level and they can alter the interfacial interaction, load transfer mechanisms, and the total composites properties [8,9]. Although great progress in numerical simulations and experimental testing has been achieved, Bahtiri et al. found a drawback that is represented by the error associated with numerical and experimental results, especially for complex loading conditions and varying environmental conditions like temperature and moisture [10].

For instance, adding graphene to CFRP has greatly increased its tensile strength, modulus, and fracture toughness [10,11]. Alexopoulos et al. found that the tensile strength is increased by as much as 30 % and

* Corresponding author.

E-mail address: md.islam@me.kuet.ac.bd (Md.A. Islam).

<https://doi.org/10.1016/j.matdes.2025.114537>

Received 5 June 2025; Received in revised form 5 August 2025; Accepted 6 August 2025

Available online 7 August 2025

0264-1275/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

fracture toughness by 25 % because of the better load carrying, crack bridging, and crack deflection mechanism [12]. Jiang et al. observed through experimental study that the improvement of different properties was found to increase at different wt% of graphene [13].

Another instance can be Carbon nanotubes (CNTs), which have excellent mechanical properties, high aspect ratio, and electrical conductivity and have also shown outstanding potential as reinforcing agents in CFRP composites. In the studies by Monfred et al. and Sulaiman et al., it has been revealed that CNTs can also improve compressive and flexural strength and electrical conductivity as well, which predisposes them to multifunctional applications [14,15]. Zakaria et al. (2017) compared CNT and GnP and reported that GNP possessed higher thermal and dielectric properties, while its mechanical properties were somewhat inferior to those of CNT [16].

In contrast, Alsaadi et al. found that silicone nanoparticles are commonly used for the enhancement of the fracture resistance and fatigue performance of CFRP composites [17]. De Pours et al. have revealed that the fatigue life of CFRP composites can be extended by approximately 50 % due to the good functioning of silica nanoparticles, which can relieve stress concentration and suppress crack growth [18]. Eskizeybek et al. found that the incorporation of CaCO₃ provided multiple toughening mechanisms, including crack pinning, crack deflection, microparticle debonding, and asperity–pore interaction etc., which enhance the toughness [19]. The incorporation of nanoparticles has also been shown to enhance the mechanical properties of glass fiber-reinforced polymer (GFRP) composites. Srinivasa Perumal et al. reported that adding 15 wt% rutile (TiO₂) nanoparticles to GFRP composites increased tensile strength by 58.3 % and flexural strength by 77.2 % [20]. Incorporating zircon (ZrSiO₄) and TiO₂ nanoparticles into GFRP composites has shown significant mechanical improvements. At 5:5 wt% loading, tensile strength increased by 74.3 %, flexural strength by 33.5 %, and energy absorption by 61.1 % compared to neat GFRP, due to better dispersion and fiber–matrix bonding. [21].

In a study by Karthik et al., it was mentioned that the complex interdependencies between the material constituents, processing parameters, and resulting mechanical properties of nanomodified composites make their behavior difficult to predict [22]. Several studies highlighted that uniform dispersion of nanoparticles as CNTs in CFRP composites is still a significant challenge because of the van der Waals-induced agglomeration, which influences the material consistency and property prediction [23–25]. With an increase in CNT content, mechanical properties initially enhance and then deteriorate because of the agglomeration of fillers and owing to weaker interfacial bonding [26]. For the reinforcement, the proper dispersion and a strong polymer–filler bonding are required; however, high contents prevent achieving these two together [27,28]. This latter factor introduces complex model mechanics due to the non-uniform dispersion of the filler (particularly under nonlinear stress–strain conditions) [27], while the scarcity of simulation studies on agglomeration highlights the demand for more advanced simulations when experimental characterization is expensive [22,28,29].

Costs, the time required, and technical barriers hinder experimental studies, which restrict nanoscale characterization [22,30]. Experimental instruments are unable to provide a complete description of the impact of aspect ratio variations or molecular scale phenomena becoming more dependent on simulations [31,32]. Al-Kharusi et al. showed that there exist large errors in AFM and TEM data, making the determination of properties more difficult [33], particularly in the interphase [34]. Chwal et al. suggested that this gap needs to be filled with computational tools [22].

Analytical models fail because of assumptions that do not hold at the nanoscale, such as the rule of mixtures and continuum approximations [35,36]. At high particle loadings, ignoring agglomeration in analytical modeling frequently leads to incorrect predictions [37–40]. Modified models enhance the precision by considering interphase and agglomeration [41,42], but difficulties persist in the irregular

configuration and multiscale composites [43,44].

While more sophisticated, Uthale et al. mentioned in a study that these numerical models are relatively challenged when modeling at the nanoscale in a continuum-based framework [45]. While finite element methods are able to be discussed, they pose a problem of computing the MD-based interphase properties because they are computationally expensive [27,46]. Liu et al. mentioned that one possible line of reasoning is that single-scale methods will have to face multiscale behaviors, anisotropy, and nonlinearities [36], and Uthale et al. mentioned that multiscale RVE approaches are necessary to connect nano to macro scales [45].

The hierarchical composite structures add complexity to the modeling. Multiple studies found that ply orientation and geometry obstruct macroscale prediction [47]. According to Yang et al., this heterogeneity is difficult to describe by means of MD simulations, and it results in unreliable predictions of strength [48]. Liu et al. mentioned that the use of simplified assumptions hinders classical analysis techniques and is not effective when dealing with multiscale phenomena [36]. According to Uthale et al., multidisciplinary multiscale approaches integrating experiments, modeling, and simulation are critical for unraveling the mechanics of CFRP nanocomposites [45].

These constraints make machine learning (ML) techniques a more effective and precise alternative to predict the mechanical properties of composites.

Machine learning (ML) for prediction and optimization of mechanical properties, a substitute for the traditional expensive and time-consuming approaches [49,50], and screen of carbon fiber reinforced polymer (CFRP) composites for process–structure–property correlations [51]. Supervised learning methods (Kernel Ridge Regression model (KRR), Gaussian Process Regression model (GPR), Support Vector Machines (SVM), decision trees, K-Nearest Neighbors (KNN), gradient boosting can predict mechanical properties with excellent performance; regression tree models predict properties like elastic moduli, shear moduli from finite element model data [52–55].

According to studies by Malashinn et al. and Zhao et al., for CFRP composites, ML is used for the optimization of stacking sequences and fiber directions; and XGBoost and gradient boosting has been reported optimum for tensile strength; decision trees, K-Nearest Neighbors (KNN) and Random Forest were found most suitable for bending strength [56,57]. In the study of Rohatgi et al. (2022), nanoparticle reinforced composites, mechanical properties were predicted by ML as in graphene-reinforced copper composites, the yield and the tensile strength were predicted by Random Forest, Artificial Neural Network, Gradient Boosting Machine, and K-Nearest Neighbor [58] and in the study of Huang et al. (2023) magnesium matrix composites the yield strength was predicted by XGBoost [59] with an R² value of 0.94 status [60,61]. Both Sharma et al. and Deb et al. mentioned in separate studies that decision tree models connect the properties of carbon fibers to the macroscopic properties of CFRPs and predict the properties developed for various reinforced plastics and polymer composites [62,63].

DTs are currently utilized for wide prediction of composite mechanical properties, relating properties of carbon fibers with macroscopic properties of CFRP [62,64] and predicting the mechanical property properties of carbon fiber-reinforced plastics and alumina polymer composites [63]. Random Forest (RF) models were also used to predict the yield and tensile strengths of graphene-reinforced copper composites [58,65], providing their usefulness for fewer data points [66,67]. According to Wang et al., Karamov et al., and Sheikh et al., the influence of gradient boosting algorithms has been remarkable as (i) Models using XGBoost outperformed to predict tensile strength [68], (ii) gradient boosting decision tree algorithms performed outstandingly well in predicting the fracture toughness (with mean squared errors less than 10 %, than ANNs with less data) [65,66]. Liu et al. (2022) applied XGBoost for the investigation of dielectric properties of epoxy with inorganic hybrid nanofiller, which outperformed RF/KNN [69]. Uddin et al. (2024) developed the use of the XGBoost model to predict

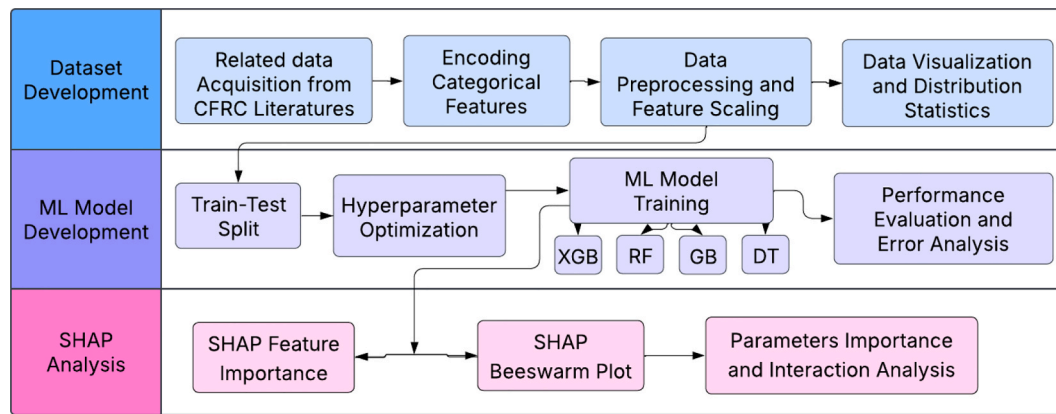


Fig. 1. Workflow of the study.

the dielectric properties of nano-filler-reinforced epoxy composites [70].

Numerous studies have found that tree-based ensemble approaches are the consistently best performers for the prediction of the mechanical properties in nanoparticle-reinforced CF composites [71–74]. The quality/ quantity of data also has an impact on the performance. According to Yilmaz et al., for the geopolymers composites, specific characteristics resulted in the strong performance of XGBoost [75]. The algorithm selection should also match the data, as decision tree models reliably predict properties on multiple composites; however, according to Deb et al., Random Forest regression exhibits greater accuracy on others [63]. Meanwhile, Che et al. found the precision and R^2 of multiple regression were 0.92 and 0.96, respectively [60].

The selection of ML models in composite research depends on dataset volume, complexity, and the specific problem of interest. It is well documented in the literature that XGBoost is the most accurate and powerful algorithm to predict composite properties, particularly with large and complex datasets [69,70]. Another study by Lim et al. showed that random Forest is known for being interpretable and noise resistant and is well suited to feature selection problems [76]. Although Gradient boost (GB) and Decision Tree(DT) perform well for small datasets, they demand skilled parameter selection for good performance, according to Adithiyaa et al. and Rajiv et al. [77,78].

Although ML models have shown remarkable efficiency in predicting the mechanical properties of composites, their “black-box” characteristic normally limits their large application, especially in safety-critical areas. This led to the creation of Explainable Artificial Intelligence (XAI) as a domain dealing with the development of methods and tools for making ML models transparent and interpretable [79]. According to Daghighi et al. and Yossef et al., XAI approaches, such as SHapley additive explanations (SHAP) [79], local interpretable model-agnostic explanations (LIME), and partial dependency plots (PDPs), can be used to gain an understanding of how ML models make decisions, such as which input features impact ML model predictions [80]. These methods are also essential for establishing trust in ML models, finding potential biases, and ensuring that the model actually learns to make predictions from useful relationships in the data instead of spurious correlations.

Hematibahar et al. (2024) employed SHAP to explain compressive strength predictions of concrete and found that silica ash and cement content played a dominant role in the compressive strength [81]. Zheng et al. (2023) LIME and SHAP were used on demand to investigate the fluctuations of tensile strength of cellular concrete, yielding important information about the effects of porosity and aggregate size on tensile strength [82]. However, according to Pande et al., limited effort has been proposed for the implementation of XAI methodology for nano-modified CFRP composites. The majority of prior work has been dedicated to traditional composites or applied only one XAI method to a

single ML model, failing to recognize that explanations may vary among different algorithms and that feature interactions should be fully understood [83]. Furthermore, the practical deployment of real-time predictive analysis and visualization of XAI results also suffers from the lack of investigation, which makes the access and application of XAI technology a difficulty.

Previous research on the mechanical properties of carbon fiber-reinforced, epoxy-based nanoparticle composites has often been constrained by limited datasets and a narrow range of manufacturing parameters. These limitations can result in predictive models that fail to represent the complex interactions influencing mechanical performance accurately. Furthermore, the lack of diversity in applied machine learning techniques highlights the need for more comprehensive and robust modeling approaches.

While previous studies have explored the use of machine learning to predict the mechanical behavior of carbon fiber/epoxy composites, many have been limited either by small datasets or by a narrow range of processing parameters. In most cases, the focus has remained on prediction accuracy, without providing insight into how specific material or manufacturing features affect performance. To address these limitations, this study aims to present a more comprehensive and diverse dataset compiled from different experimental sources, covering a wide range of processing conditions and particle types. Furthermore, this work intends to integrate ensemble ML models with explainable tools such as SHAP to not only improve prediction accuracy but also provide clear understanding of feature influence. The use of SHAP allows for identifying the most impactful processing parameters, which can guide optimized composite design, reduce material waste, and lower experimental time and cost. By combining a large, diverse dataset with interpretable machine learning, this study offers a novel framework for the accurate and transparent prediction of tensile and flexural strength in nanoparticle-reinforced CF/epoxy composites, with practical relevance to material optimization and performance-based design in composite manufacturing.

2. Methodology

From Fig. 1, it can be seen that the research includes a three-phase workflow. The first is dataset development, which involves acquiring data from the literature, encoding categorical features, preprocessing, scaling, and visualizing the data. In ML Model Development, data is split for training and testing. Hyperparameter optimization refines XGBoost, Random Forest, Gradient Boosting, and Decision Tree models before final training and performance evaluation. Lastly, SHAP Analysis interprets model predictions by assessing feature importance, generating beeswarm plots, and analyzing parameter interactions to provide insights into composite behavior.

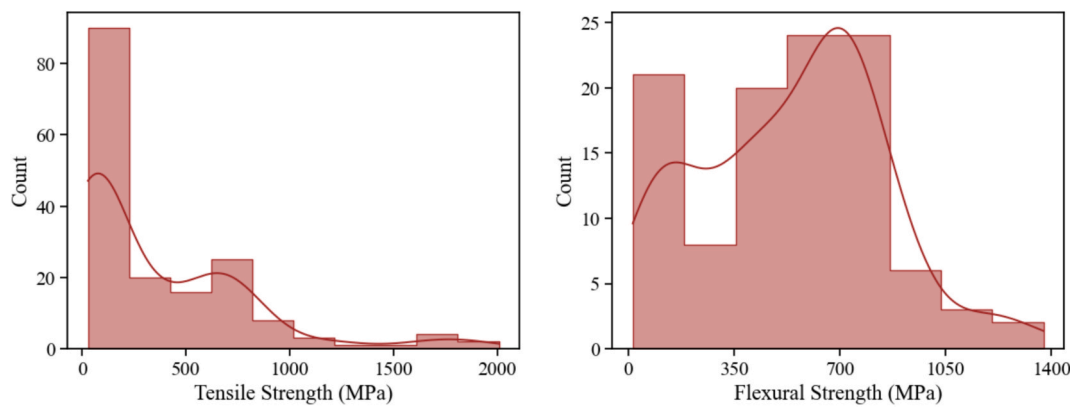


Fig. 2. Distribution of number of experiments with respect to (a) Tensile dataset and (b) Flexural dataset.

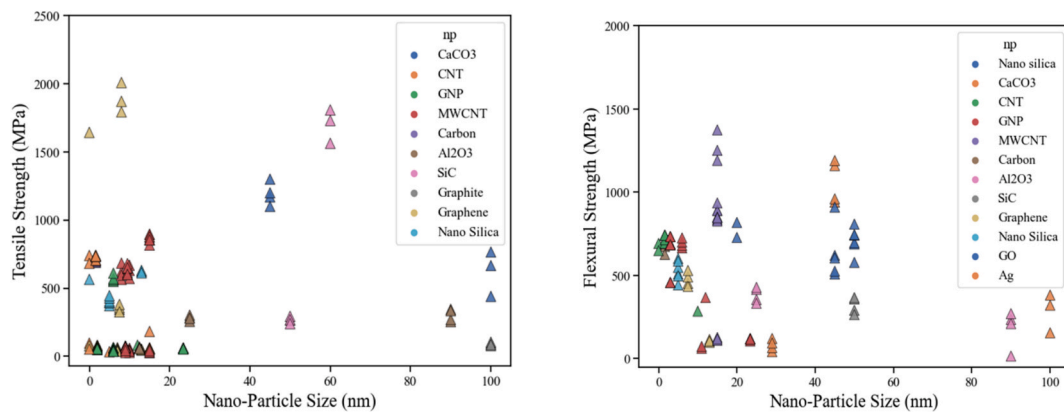


Fig. 3. Distribution of particle size with respect to (a) Tensile strength and (b) Flexural strength.

2.1. Dataset Development

The dataset comprises 174 experimental results for tensile strength and 108 experimental results for flexural strength collected from 50 Carbon fiber and epoxy-based nanoparticle filler publications [1–6,10,12–14,17,18,22,24,89–118]. The data has been collected from both journal articles and conference proceedings with the distribution of number of experiments as is shown in Fig. 2. In the compiled dataset, carbon fiber density ranged from 1.5 to 3.05 g/cc, and epoxy matrix density ranged from 1.1 to 1.33 g/cc. However, since this information was not consistently reported across all sources, these two features were excluded from model training and testing.

Most publications provided data in numerical form, which was directly extracted and included. In cases where data were presented graphically in bar charts or scatter plots, they were digitized using an online tool called Web Plot Digitizer, which enables extraction from visual plots through interpolation and extrapolation [119]. Several challenges were encountered during data collection, particularly missing or incomplete information. When particle size values were not explicitly mentioned in the articles, they were obtained from the manufacturer's technical data sheets referenced in the literature. However, if any of the processing parameters, such as sonication temperature, curing duration, post-curing temperature, post-curing duration, or fiber orientation, were entirely absent, the corresponding data rows were excluded to avoid introducing speculative values. It is to be noted that categorical input features like orientation, manufacturing process, and particle type were encoded to numerical values. To prepare the dataset for ML models, these categorical features were transformed into a numerical format using Label Encoding with the sklearn's LabelEncoder module [120]. This method assigns a unique integer to each distinct

category in a feature. Since some publications did not report flexural properties, two separate datasets were created. Lastly, to facilitate efficient model training and testing, the input features were standardized using the StandardScaler from sklearn.preprocessing [121–124]. The target variable and the encoded categorical features were excluded from scaling to preserve their interpretability. Both datasets included three types of fiber orientation: Unidirectional, Woven, and Cross-ply, and also identified eight distinct manufacturing processes across the tensile and flexural data. These processes, listed from most to least common, are: Hand Layup, Vacuum Bagging, Vacuum-Assisted Resin Transfer Molding (VARTM), Dispersion, Filament Winding, Hot Pressing, Plasma Treatment, and the Dispersion Method. Table 2 depicts the statistical summary of the overall dataset for the numerical features for both tensile and flexural tests. Fig. 3 shows the types of particles that were used for the tensile and flexural datasets, respectively. From Fig. 3(a), it can be seen that most of the tensile strength lies between 27.5 MPa and 1000 MPa, and only a few composite particles gained extraordinarily high tensile strength, reaching close to 2009 MPa. It is also noticeable that carbon fiber reinforced with graphene particles has gained the highest tensile strength with a particle size of 8 nm. The flexural distribution shown in Fig. 3(b), shows almost a bell curve, with most of the composites achieving bending strength with a centralized cluster within 350 to 1050 MPa. Here, the multi-walled carbon nanotube has the highest flexural strength of 1374.02 MPa. Both figures indicate that particles smaller than 20 nm, particularly carbon-based nanoparticles such as MWCNTs, CNTs, and graphene, consistently achieve the highest mechanical properties overall.

Table 1
Input and Output parameters of CFRC.

Input Parameter	Details	Unit
Fiber Layer	Number of carbon fiber plies used in the composite	–
Sonication Duration	Length of time the mixture is sonicated	Hours
Sonication Temp	Temperature is maintained during the sonication process	Celsius
Curing Duration	Length of time the composite is held for curing	
Curing Temp	Temperature at which the primary curing of the matrix occurs	Celsius
Postcuring Duration	Length of time the composite undergoes postcuring heat treatment	Hours
Postcuring Temp	Temperature used for the post-curing heat treatment	Celsius
Particle Size	Size of the reinforcing nanoparticles	nm
Particle Ratio	Proportion of nanoparticles in the composite	%
Fiber Orientation	Angle of the carbon fibers within the layers are aligned.	–
Manufacturing Process	Specific method used to fabricate the composite	–
Particle Type	Specific material or type of nanoparticle used	–
Output Parameter		
Tensile Strength	Maximum stress before breaking under tension.	MPa
Flexural Strength	Maximum stress before yielding under bending.	MPa

2.2. Model development

For the prediction of the mechanical properties of the carbon fiber-reinforced epoxy-based nanoparticle composites, several regression algorithms were evaluated. The models assessed included XGBoost (XGB), Random Forest (RF), Gradient Boosting (GB), and Decision Tree (DT).

These models were chosen due to their effectiveness in medium-sized datasets and their ability to handle nonlinear interactions. To optimize model performance, hyperparameters for all machine learning models were selected using GroupKFold from the Scikit-learn library [84]. This method searched the the groups parameter was set to an array of source paper identifiers (paper ids), ensuring that all data points from a single paper were assigned to the same fold during cross-validation. This group-based CV strategy prevented data leakage by maintaining the independence of training and test sets across the 50 source papers. The train-test ratio was kept at 80:20. After that, the models were trained and tested on the 12 input features detailed in Table 1, using the best parameters obtained from the grid search.

2.2.1. Random forest

Random Forest (RF) is a powerful supervised machine learning algorithm that utilizes the collective intelligence of multiple decision trees to enhance predictive accuracy. It works on the principle of ensemble learning, which is based on the bootstrap aggregating (bagging) method. In this approach, numerous decision trees are independently constructed by sampling random subsets of the training data with replacement

(bootstrap samples). Randomness is further introduced by considering only a random subset of features at each split, which helps to correlate trees and improve generalization by reducing variance. The RF algorithm functions in two primary stages. Firstly, a forest of decision trees is built by training each tree on its respective bootstrap sample with random feature selection. Secondly, each tree in the forest predicts a given input. The final predicted output is obtained for regression tasks by averaging the predictions from all the individual decision trees in the forest. The following equation represents this averaging process:

$$\hat{Y} = \frac{1}{B} \sum_{b=1}^B Y_b(X') \quad (1)$$

here Y is the final predicted output for the input X' , $Y_b(X')$ is the prediction of the b -th decision tree for the input X' , and B is the total number of trees in the forest. Increasing the number of trees (B) generally leads to improved accuracy and reduced variance in the predictions due to the effect of averaging [85].

For random forest regression, $n_estimators$ was kept at 100 and 250 for tensile and flexural strength models, respectively. The rest of the parameters were kept at the default.

2.2.2. XGBoost

In the current study, XGBoost (Extreme Gradient Boosting) is applied, which is a supervised machine learning algorithm and uses boosting techniques. It combines multiple weak learning models to construct a strong prediction model and builds an ensemble of decision trees in a sequential manner, where each subsequent tree focuses on minimizing the residual errors of its predecessor trees. The algorithm optimizes an objective function that combines a loss function $L(y_i, \hat{y}_i)$ for error minimization and a regularization term $\Omega(f_k)$ to prevent overfitting. The objective function can be expressed as:

$$\text{Objective} = \sum_{i=1}^n L(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (2)$$

where K is the number of trees, f_k is the prediction function for the k -th tree, y_i is the observed value, and $\hat{y}_i$ is the predicted value [86].

In this study, the XGBoost model was developed with the number of trees set to 100, the learning rate of 0.1, and the leaf node set to zero for both tensile and flexural models. The alpha was kept at 10 for the tensile model and 1 for the flexural model. The rest of the parameters were set to default.

2.2.3. Gradient boosting

Gradient Boosting (GB) is another powerful ensemble machine-learning technique that builds a strong predictive model by sequentially adding weak learners. Unlike bagging methods li, Gradient Boosting builds trees in a step-wise manner, with each new tree trained to predict the residuals of the preceding ensemble. The algorithm uses gradient descent to minimize a chosen loss function $L(y_i, \hat{y}_i)$ by iteratively adjusting the model. At each iteration m , the algorithm adds a new tree $h_m(x)$ to the model to reduce the residuals $r_{i,m}$:

Table 2
Statistical distribution of the tensile and flexural dataset.

Parameter	Tensile Dataset				Flexural Dataset			
	Min	Max	Mean	Std	Min	Max	Mean	Std
Fiber Layer	0	16	7.55	5.84	3.0	24.0	10.22	5.76
Sonication Duration	0	3	0.68	0.48	0.0	6.00	1.02	1.39
Sonication Temp	20.0	120	31.47	21.18	20	120.00	32.26	24.18
Curing Duration	0.50	24.00	12.65	10.63	0.5	24.00	12.58	11.07
Curing Temp	20.0	150.0	56.90	40.33	20.0	150.00	62.55	48.35
Postcuring Duration	0	24.0	7.17	9.48	0	24.00	2.61	5.07
Postcuring Temp	0	700.0	82.12	121.86	0	700.00	82.87	147.72
Particle Size	0	100	17.49	23.98	0	100	23.06	24.98
Particle Ratio	0	10.0	1.26	1.53	0	20	1.50	2.57
Strength	27.5	2009.0	366.87	427.38	15.0	1374.02	540.83	303.20

Table 3
Performance of Models for Tensile strength and Flexural strength.

	Model	Training stage			Testing stage		
		R ²	MAE	RMSE	R ²	MAE	RMSE
tensile test	XGB	0.999	0.634	2.767	0.980	30.452	64.208
	RF	0.990	19.080	41.133	0.973	40.786	75.504
	GB	0.998	11.093	17.219	0.975	42.567	73.340
	DT	0.999	4.058	9.844	0.983	28.192	59.312
Flexural test	XGB	0.999	1.335	8.463	0.928	57.022	86.194
	RF	0.982	27.076	39.627	0.929	61.407	84.740
	GB	0.991	20.845	28.388	0.931	59.964	83.828
	DT	0.999	2.705	9.639	0.873	73.613	114.048

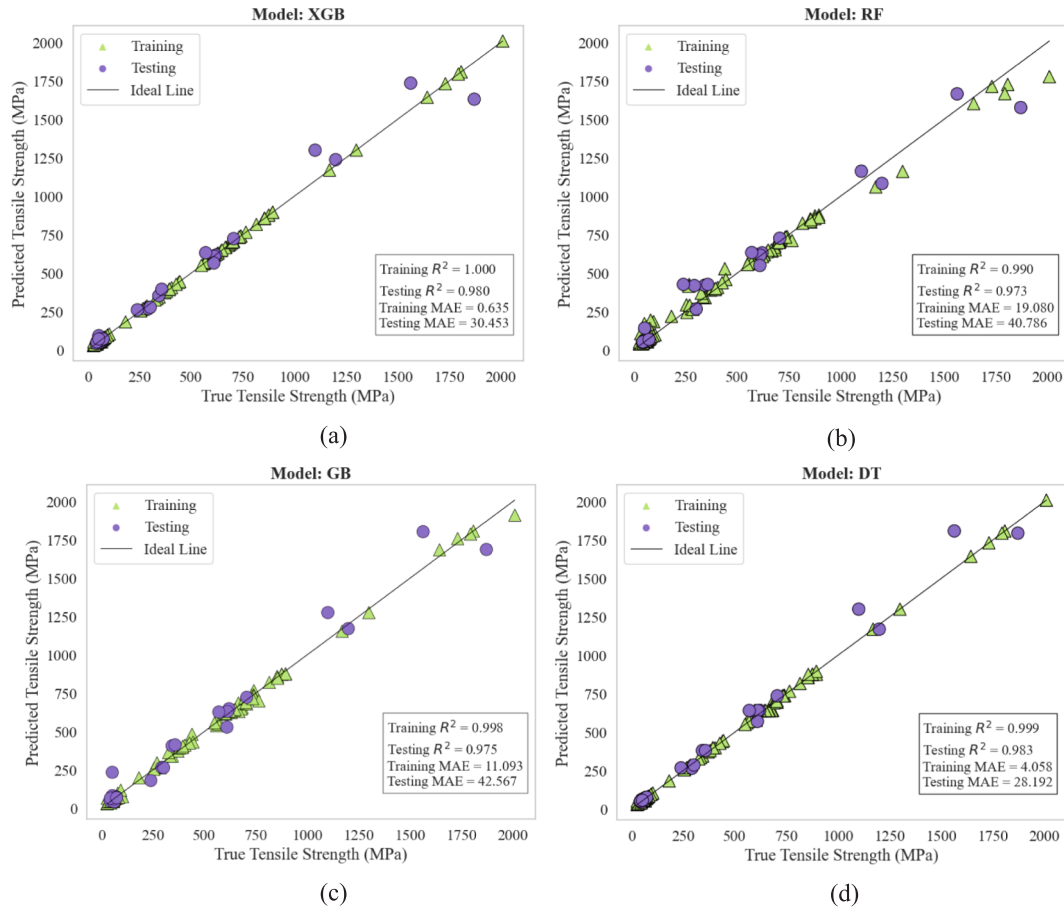


Fig. 4. Training and testing performance of (a) XGB, (b) RF, (c) GB and (d) DT model for Tensile strength prediction.

$$r_{i,m} = -\frac{\partial L(\mathbf{y}_i, \hat{\mathbf{y}}_{i,m} - 1)}{\partial \hat{\mathbf{y}}_{i,m-1}} \quad (3)$$

A new weak learner is fit to the negative gradient of the loss function concerning the current model's predictions. The predictions of these sequentially trained trees are then combined to form the final prediction [87]. For this study, the tensile strength model was developed with a learning rate of 0.3, using the 'squared_error' loss function, and setting the number of estimators to 100. For flexural strength prediction, the model used a learning rate of 0.5, the 'huber' loss function, and 150 estimators.

2.2.4. Decision tree

A Decision Tree (DT) is a non-parametric supervised learning algorithm for regression that repeatedly splits the data based on the values of its features, forming a tree structure. Internal nodes test features,

branches represent outcomes, and leaf nodes give the final prediction, typically the average of target values in that leaf. Tree building selects splits based on criteria like Mean Squared Error (MSE) to minimize variance in subsets. The process stops based on criteria like max depth or min samples per leaf [88].

In this study, the DT Regressor for tensile strength was configured with a maximum depth of None and a minimum of 1 sample per leaf, while the flexural strength regressor used a maximum depth of 10 and a minimum of 1 sample per leaf. All other parameters for both regressors were kept at their default values.

3. Results and discussions

3.1. Tensile strength prediction

The tensile strength (MPa) prediction for CFRP composites was assessed using four machine learning models: XGB, RF, GB, and DT.

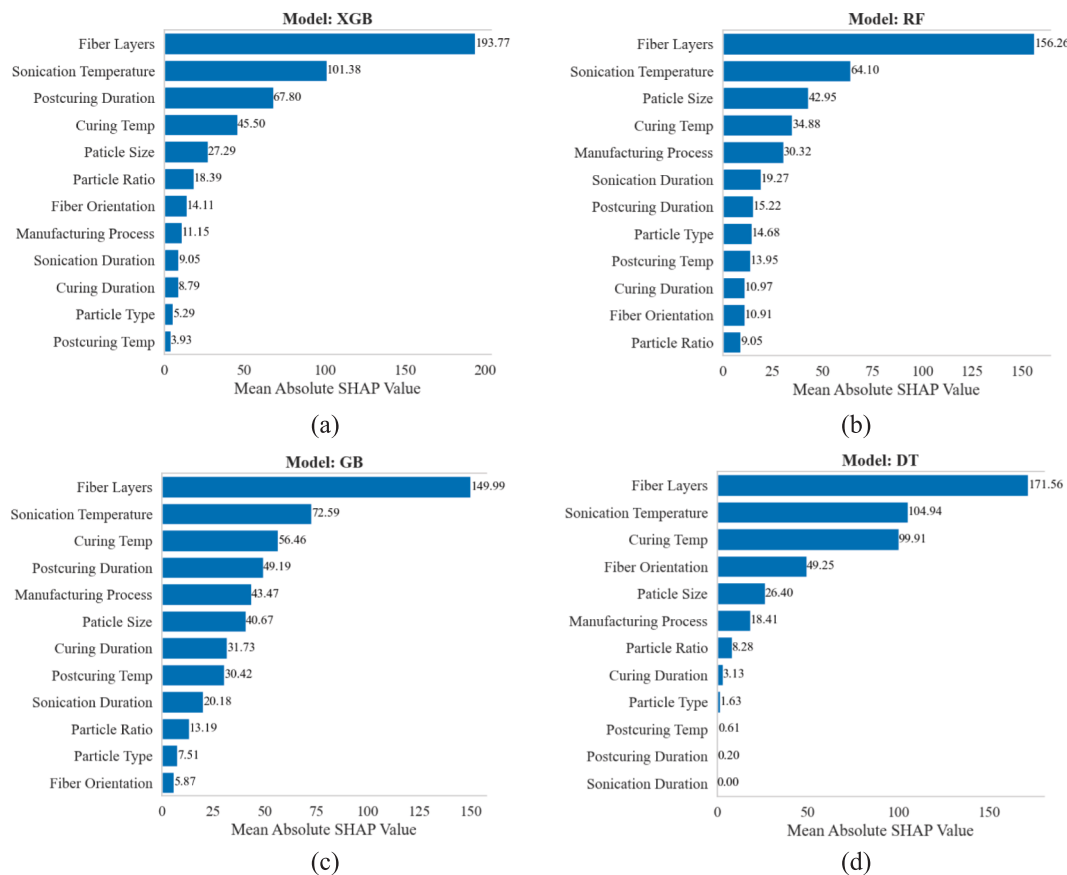


Fig. 5. Mean absolute SHAP (global) explanations obtained for tensile strength models (a) XGB, (b) RF, (c) GB, and (d) DT model.

Results, shown in Table 3, highlight the models' predictive capabilities across training and testing phases, providing insights into their applicability for composite material analysis. In Fig. 4a, it is observed that the XGB model has shown the highest performance during the training phase, with an R^2 value of 0.999, MAE of 0.634 MPa, and the least RMSE of 2.767 MPa, indicating an almost perfect fit. In the testing phase, the XGB model maintained robust generalization capability, achieving an R^2 of 0.980, with moderate increases in MAE (30.452 MPa) and RMSE (64.208 MPa). The RF model (Fig. 4b) performed well overall but with slightly lower accuracy compared to the other ensemble methods. It has gained a training R^2 of 0.990, MAE of 19.080 MPa, and RMSE of 41.133 MPa. During testing, it achieved R^2 of 0.973, with MAE of 40.786 MPa and highest RMSE of 75.504 MPa, showing solid but slightly less precise predictions. The GB model (Fig. 4c) achieved strong training performance ($R^2 = 0.998$, MAE = 11.093 MPa, RMSE = 17.219 MPa) and superior testing results ($R^2 = 0.975$, MAE = 42.567 MPa, RMSE = 73.340 MPa). However, its error margins during testing were slightly higher than XGB model.

The DT model (Fig. 4d) achieved the highest R^2 score of 0.983 among all models in the testing phase. The training performance of the DT model is nearly perfect ($R^2 = 0.999$, RMSE = 9.844 MPa), and the model also maintains a relatively low MAE (28.192 MPa) and RMSE (59.312 MPa) in the testing phase.

3.1.1. SHAP importance for tensile strength

Based on the models' performance on tensile test data, global SHAP explanations are presented in Fig. 5, illustrating the feature importance for tensile strength prediction across four different models. All four models consistently identified fiber layers as the most dominant feature, with the XGB model (Fig. 5a) attributing the highest importance (193.77), followed by DT (Fig. 5d), RF (Fig. 5b), and GB (Fig. 5c). This highlights the significant role fiber layering plays in tensile strength.

Sonication temperature emerged as the second most important feature across all models, with values ranging from 64.10 in RF to 104.94 in DT. Curing temperature also showed strong importance across all models. It was ranked third by GB (56.46) and DT (99.91) and fourth by both XGB (45.50) and RF (34.88), indicating a common recognition of its high effect on matrix hardening and polymerization processes. Post-curing duration exhibited relatively high importance in XGB (67.80, ranked 3rd) and GB (49.19, ranked 4th), while it was less impactful in RF (15.22, 6th) and DT (0.20). Particle size ranked moderately high in XGB (27.29) and RF (42.95), with slightly lower values in GB (40.67) and DT (26.40), hinting at its moderate effect on tensile performance. Interestingly, fiber has a common recognition of its high effect on matrix hardening and polymerization processes. Post-curing duration exhibited relatively high importance in XGB (67.80, ranked 3rd) and GB (49.19, ranked 4th), while it was less impactful in RF (15.22, 6th) and DT (0.20). Particle size ranked moderately high in XGB (27.29) and RF (42.95), with slightly lower values in GB (40.67) and DT (26.40), hinting at its moderate effect on tensile performance.

Interestingly, fiber orientation held mid-to-low importance, with the highest relevance in DT (49.25) and XGB (14.11), whereas GB (5.87) and RF (10.91) considered it less critical. This may reflect varying sensitivities of models to directional properties of reinforcement. Particle ratio and particle type were considered less impactful by all models, though DT and RF assigned moderate importance to particle type (1.63 and 14.68, respectively). The manufacturing process showed moderate importance in RF (30.32) and GB (43.47), but was lower in DT and XGB. Curing duration, post-curing temperature, and sonication duration were the least influential features across models, suggesting a limited contribution of these variables in tensile strength prediction within the gathered dataset.

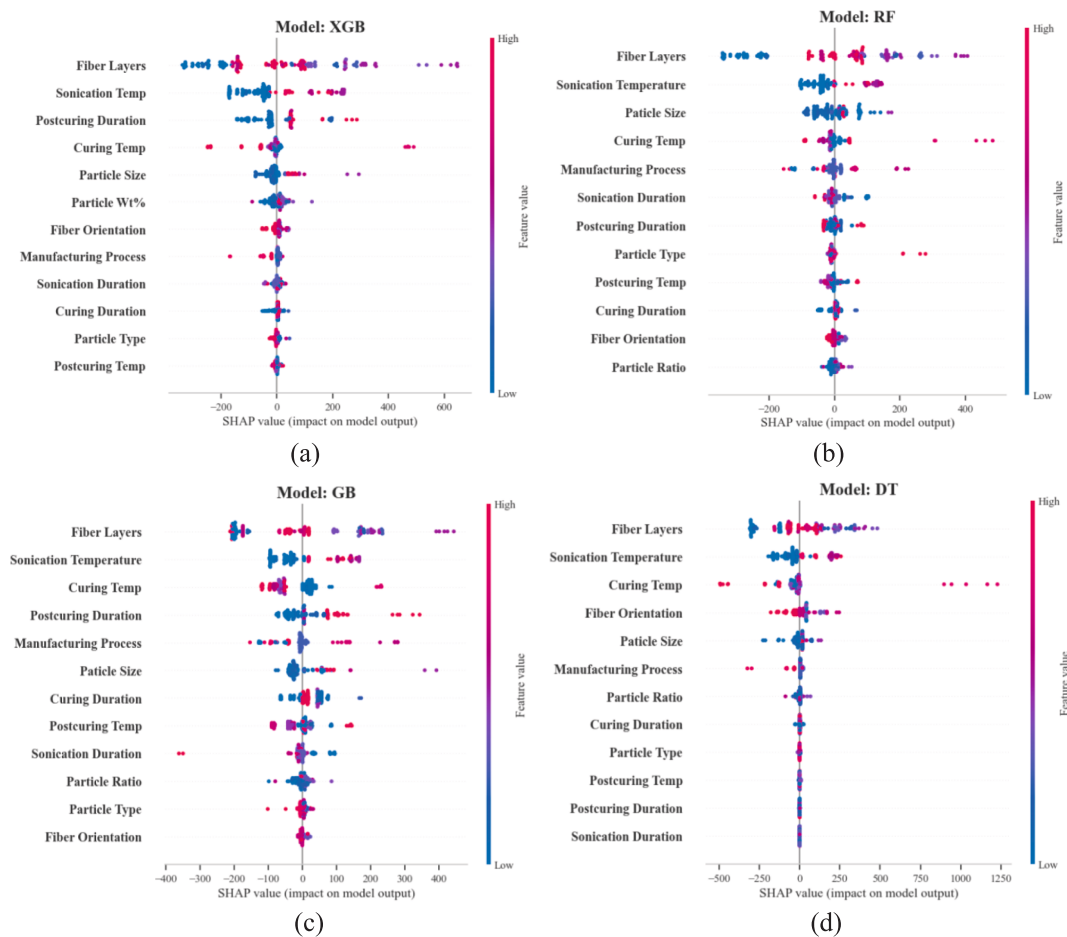


Fig. 6. SHAP beeswarm Plot for tensile models (a) XGB, (b) RF, (c) GB, and (d) DT model.

3.1.2. Beeswarm plot analysis for tensile strength

Global explanations obtained using SHAP are presented in Fig. 6, illustrating the distribution and direction of feature impact for tensile strength prediction across the four different models. The beeswarm plots show how the SHAP value for each instance is influenced by the feature's value, where blue means a lower feature value and red means a higher feature value. From all the beeswarm plots, it is visible that Fiber Layers, being the most influential feature, show that a higher number of fiber layers consistently exhibit large positive SHAP values, indicating a strong tendency to increase tensile strength. Conversely, it is also visible that fewer fiber layers lead to significant negative SHAP values and decreased tensile strength. This finding is expected as fiber layers are the primary load-bearing components in CFRP under tensile stress. Sonication Temperature emerges as a highly important feature for tensile strength. Across the models, samples processed at higher sonication temperatures generally correlate with positive SHAP values, suggesting an increase in tensile strength. Lower sonication temperatures tend to result in negative SHAP values. An appropriate sonication temperature can improve the dispersion of nanoparticles or other fillers and enhance the interfacial bonding between the fibers/fillers and the matrix, both of which are crucial for efficient stress transfer and overall tensile strength.

Post-curing Duration shows an unclear impact in the RF, GB, and DT models, but a positive impact on tensile strength, particularly in the XGB model. The model in Fig. 6a shows that shorter post-curing durations often contribute to decreased tensile strength. This is likely due to short post-curing not allowing for complete cross-linking of the polymer matrix capable of supporting the fibers under tensile load. Curing Temperature also demonstrates a generally negative impact on tensile strength in the GB model (Fig. 6c), where higher curing temperatures

tend to result in negative SHAP values. It shows that higher temperatures can lead to a decrease in tensile strength, potentially due to thermal degradation of the matrix. Although other models like RF (Fig. 6b) and DT show an indefinite impact. While the distribution of Particle size is quite scattered, there is a tendency for larger particle sizes to correlate with positive SHAP values in all models except DT (Fig. 6d). The reasons for this can vary depending on the material system, possibly by affecting how the matrix forms, how cracks spread, or how particles interact with the matrix and fibers. Other features like Particle Ratio, Fiber Orientation, Manufacturing Process, Sonication Duration, Particle Type, Postcuring Temp, and Curing Duration generally show more scattered distributions of SHAP values clustered closer to zero or with less clear, indicating a comparatively lesser impact on tensile strength within this dataset compared to the top-ranked features.

3.2. Flexural strength prediction

The predictive performance of the four machine learning models—XGBoost (XGB), Random Forest (RF), Gradient Boosting (GB), and Decision Tree (DT)—for the flexural strength of the carbon fiber reinforced composites is detailed in Table 3. In Fig. 7a, the XGB model demonstrated high performance during the training phase, achieving an R^2 value of 0.999, an MAE of 1.335, and an RMSE of 8.463, indicating an almost perfect fit to the training data. In the testing phase, the XGB model maintained the performance, achieving an R^2 of 0.928. While its errors increased significantly compared to training, it still kept the lowest MAE of 57.022 and an RMSE of 86.194 among the models, suggesting superior predictions on average for unseen data. From Fig. 7b, the RF model also showed strong performance in the training phase with

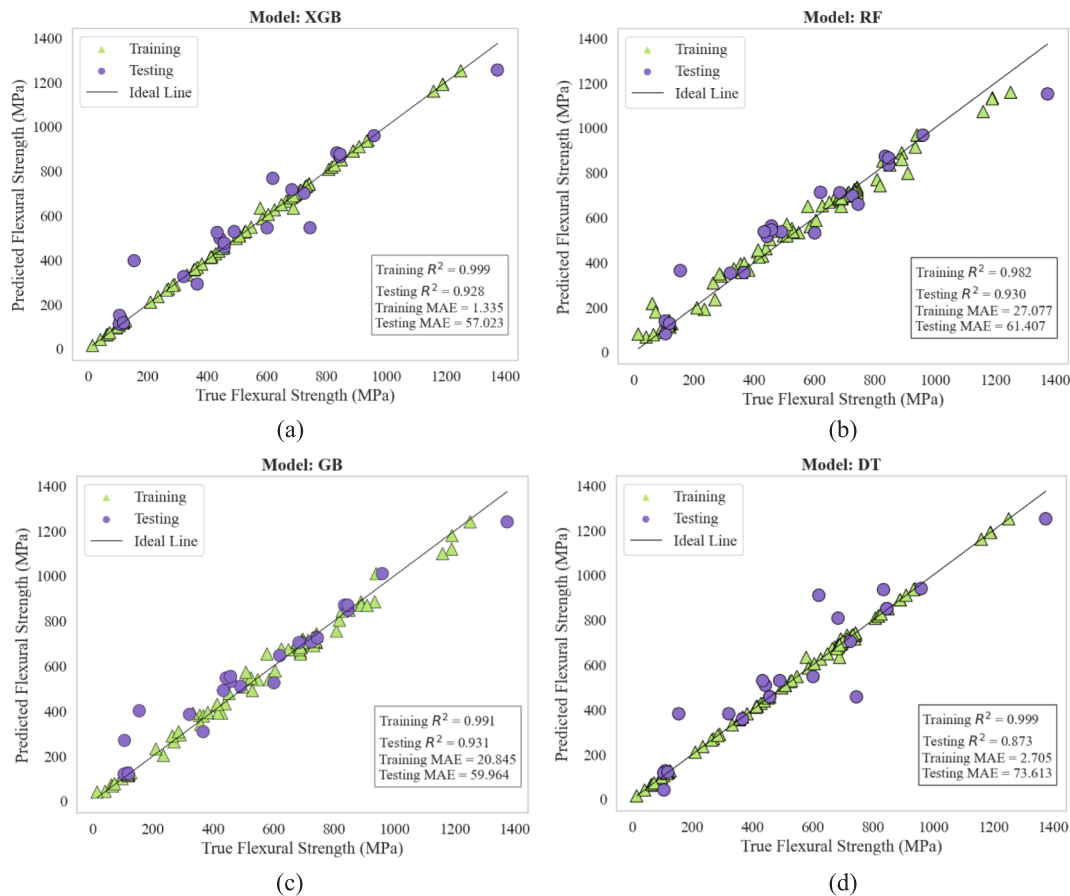


Fig. 7. Training and testing performance of (a) XGB, (b) RF, (c) GB and (d) DT model for Flexural strength prediction.

an R^2 of 0.982, MAE of 27.076, and RMSE of 39.627. On the testing set, the RF model exhibited good generalization, achieving an R^2 of 0.929, an MAE of 61.407, and an RMSE of 84.740, which is close to the XGB model's overall R^2 and RMSE.

The GB model (Fig. 7c) also achieved a high training R^2 of 0.991, MAE of 20.845, and RMSE of 28.388. In the testing phase, the GB model demonstrated slightly superior overall performance compared to RF and XGB, achieving the highest R^2 score of 0.931 and the lowest RMSE of 83.828 among all models. Its testing MAE was 59.963, further highlighting its good generalization ability in predicting flexural strength. The DT model (Fig. 7d) achieved an R^2 of 0.999, an MAE of 2.705, and an RMSE of 9.639 on training data. However, its performance on the testing set was noticeably lower compared to the rest of the models, with an R^2 of 0.873, MAE of 73.612, and RMSE of 114.048.

3.2.1. SHAP importance for flexural strength

Across all four models shown in Fig. 8, Curing Duration consistently emerges as the most dominant feature for flexural strength, exhibiting the highest mean absolute SHAP values of 75.20 in DT (Fig. 8d), 153.84 in XGB (Fig. 8a), 144.70 in GB (Fig. 8c), and 144.03 in RF (Fig. 8d). Sonication Duration is the second most important feature across all models, with mean absolute SHAP values ranging from 47.62 in RF to 108.26 in DT. Particle Type shows considerable importance in GB (55.66), XGB (39.55), and RF (39.87), and moderate importance in DT (16.38). Fiber Layers, which were the most dominant feature for tensile strength in the previous analysis, show varied importance for flexural strength across the models. It is moderately important in DT (48.82), and less so in XGB (32.97), RF (24.68), and GB (21.05). This suggests that, despite being fundamental, their relative importance for flexural strength compared to curing and sonication duration is perceived differently by these models. Curing Temperature is a significant feature across all models, with mean

absolute SHAP values ranging from 31.72 in XGB to 48.32 in GB (Fig.), 39.85 in DT, and 40.94 in RF. This reinforces the understanding that the temperature maintained during the curing process is crucial for achieving the desired matrix properties necessary for resisting bending loads. Particle Ratio shows moderate importance in XGB (30.26), DT (26.18), GB (18.00), and RF (16.03). Fiber Orientation is moderately important in DT (23.50), GB (16.47), and XGB (14.78) and less so in RF (12.25). The manufacturing process shows moderate importance in DT (24.01), RF (12.16), and GB (9.11) and less in XGB (7.58), indicating that the types of manufacturing steps have a moderate impact overall. Fiber Orientation, Manufacturing Process, Particle Size, Sonication Temperature, Postcuring Temp, and Postcuring Duration generally show lower mean absolute SHAP values across most models. This suggests a comparatively limited overall contribution to flexural strength prediction within this dataset.

3.2.2. Beeswarm plot analysis for flexural strength

From the beeswarm plot in Fig. 9, it is visible that Curing Duration has a consistent positive impact across all the models, where composite samples with lower feature values tend to have a lower flexural strength (negative SHAP values), and samples with higher feature values have better flexural strength. Except for a few data points, the overall pattern strongly indicates that longer curing duration improves flexural strength, likely because sufficient curing time allows the polymer matrix to achieve a higher degree of cross-linking, creating a more rigid and stronger network that effectively transfers stress under bending loads. Sonication Duration shows a tendency towards a negative impact on flexural strength across the models. Samples with lower sonication duration mostly exhibit positive SHAP values. This suggests that a shorter sonication period might be more impactful by avoiding adverse effects such as excessive matrix heating, which could alter the unique

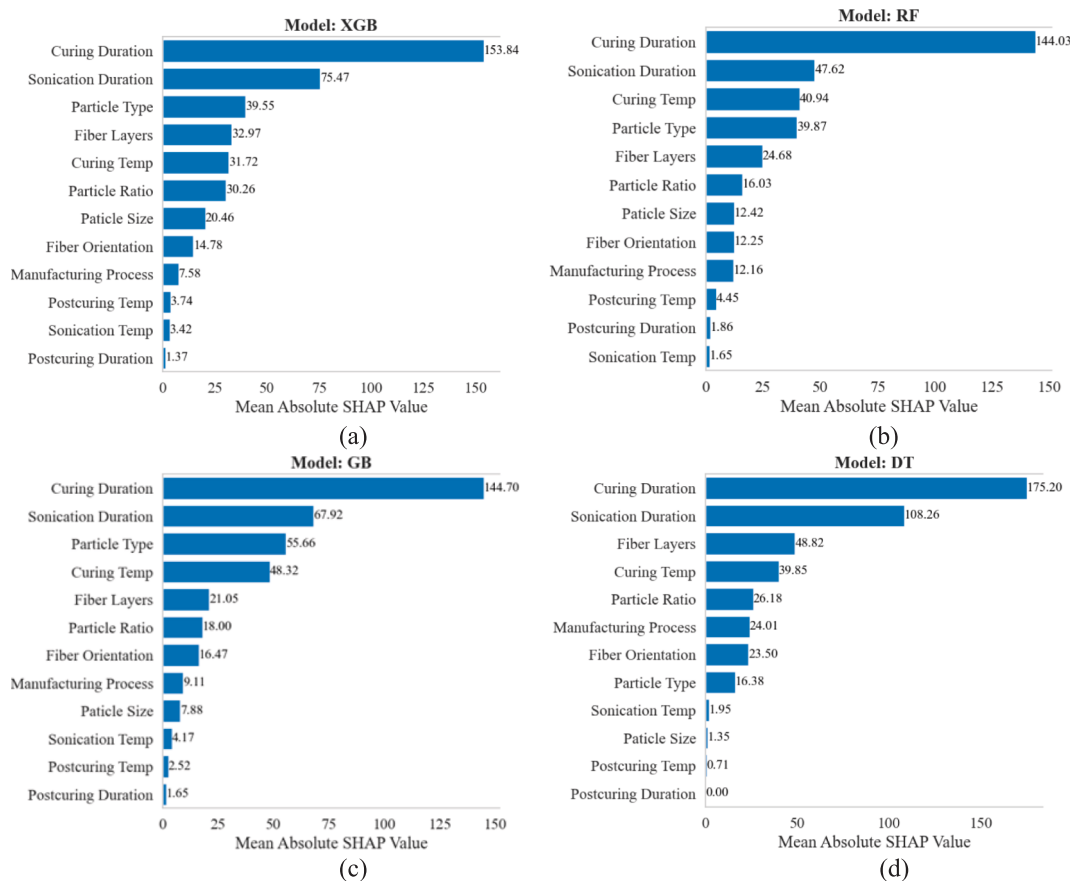


Fig. 8. Mean absolute SHAP (global) explanations obtained for tensile strength models (a) XGB, (b) RF, (c) GB and (d) DT model.

properties of nanoparticles. For Particle Type, the bee swarm plot reveals a mixed impact despite having higher mean absolute SHAP values. This variability is likely due to different particle types exhibiting distinct bonding characteristics with the fibers and matrix, despite of influencing the composite's ability to bear bending loads. For Curing Temperature, almost all the models conclude that higher curing temperatures can decrease the bending strength, referring to high curing temperatures can lead to thermal degradation of the matrix, induce excessive residual stresses due to thermal expansion mismatch, damage the fibers, or cause matrix embrittlement that decreases flexural strength.

Fiber Layers generally show a positive impact on flexural strength as the primary load-bearing component. However, the beeswarm plot for flexural strength doesn't show a clear positive correlation across all models. In all the models, instances with both lower and higher numbers of layers can result in increased or decreased flexural strength. It suggests that the impact of fiber layers does not solely depend on the number of layers but is also influenced by the matrix properties, the quality of the fiber-matrix interface (potentially affected by sonication and particle dispersion), and how well the layers are integrated during the manufacturing process and type of particles complicating the direct relationship between layer count and flexural strength. Other features like Particle Ratio, Fiber Orientation, Manufacturing Process, Particle Size, Sonication Temperature, post-curing Temperature, and post-curing Duration generally show more scattered distributions of SHAP values clustered closer to zero or with less clear positive or negative trends across the models, indicating mixed impact on flexural strength within this dataset.

3.3. Limitations of the study

Despite the robust performance of the models, it is important to

acknowledge the limitations of this study. The comprehensive dataset was aggregated from 50 different peer-reviewed articles, which may introduce variability from unrecorded differences in experimental procedures across laboratories. This could result in a form of dataset bias. Furthermore, the models were trained specifically on carbon fiber/epoxy composites modified with a specific set of nanoparticles. Therefore, the direct application of these models to other types of composites (e.g., those with different fiber or matrix materials) would require further training and validation. The exclusion of certain features like carbon fiber and epoxy matrix density due to their absence in many source publications also represents a constraint on the model's predictive scope. Future work should aim to expand the dataset with more controlled experimental data and a wider range of material systems to enhance the models' generalizability.

4. Conclusion

This study presents four explainable machine learning (ML) models—XGBoost (XGB), Random Forest (RF), Gradient Boosting (GB), and Decision Tree (DT)—to predict the tensile and flexural strength of carbon fiber/epoxy (CF/epoxy) composites reinforced with nanoparticles. The models were developed using a compiled dataset of 174 tensile and 108 flexural experimental results from 50 published literature. A total of 12 input parameters, including composite design and processing features such as particle size, particle type, sonication conditions, curing schedules, and fiber orientation, were considered. To ensure model generalization and reduce overfitting, all models were trained and validated using 5-fold cross-validation. SHapley Additive exPlanations (SHAP) were employed for model interpretability. The major findings of this study are summarized as follows:

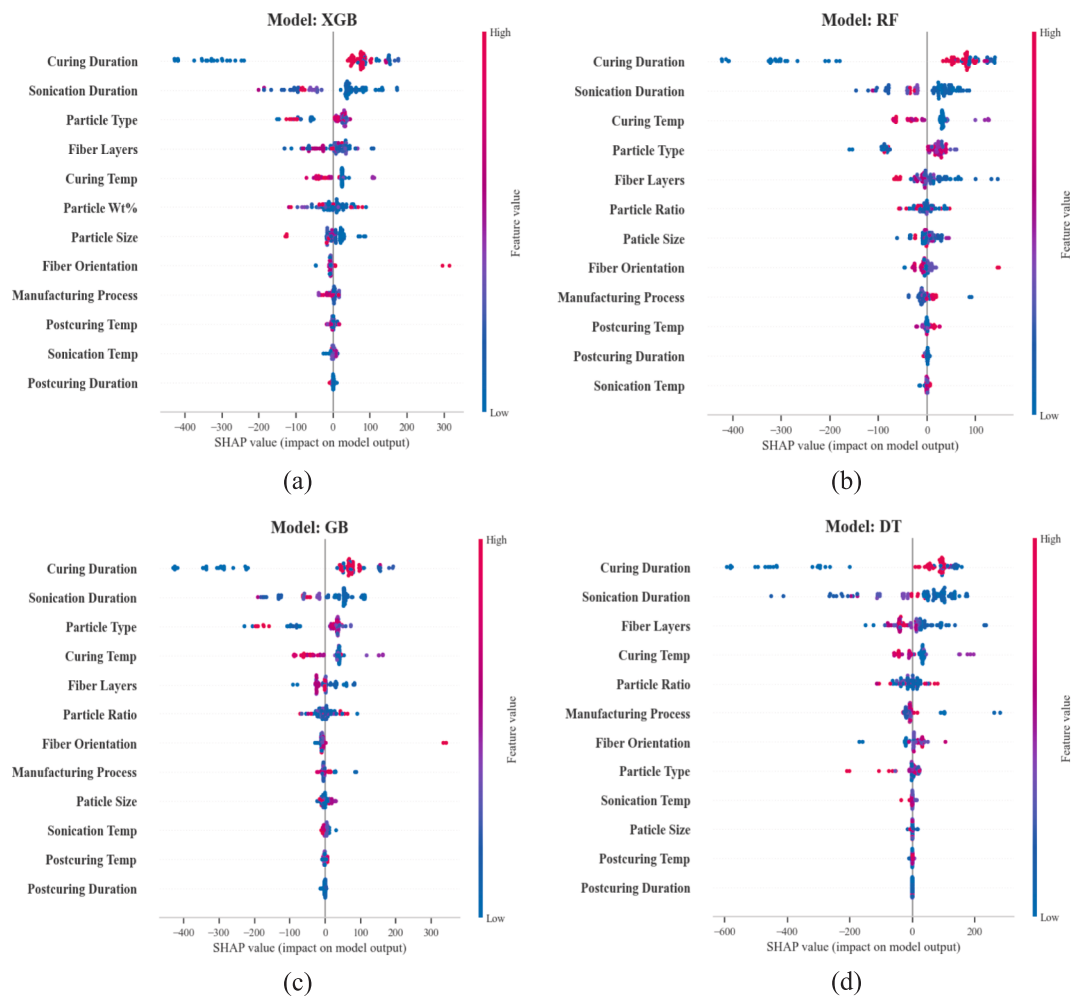


Fig. 9. SHAP beeswarm Plot for Flexural models (a) XGB, (b) RF, (c) GB and (d) DT model.

- The study confirms that carbon-based nanoparticles such as graphene and multi-walled carbon nanotubes (MWCNTs), particularly those under 20 nm, significantly enhance mechanical performance. Graphene-reinforced composites achieved tensile strength up to 2009 MPa, while MWCNTs yielded flexural strength up to 1374 MPa.
- A comparative evaluation of model performance revealed that Decision Tree achieved the highest test R^2 of 0.983 for tensile strength prediction, while Gradient Boosting achieved the highest test R^2 of 0.931 for flexural strength prediction. Notably, the XGBoost model showed highly consistent results across both training and testing datasets, achieving a test R^2 of 0.980 for tensile and 0.928 for flexural strength.
- SHAP-based feature importance analysis revealed that the most critical feature for tensile strength prediction was the number of fiber layers, followed by sonication temperature and curing temperature. The plots also revealed how curing and sonication duration have less impact on tensile strength. The importance plots for flexural strength identified curing duration as the most influential parameter, followed by sonication duration and particle type. In contrast, post-curing duration and sonication temperature had minimal impact on flexural strength enhancement.
- The SHAP beeswarm plots offered valuable insights into feature interactions, illustrating both the direction and magnitude of each parameter's effect. Notably, the plots revealed that higher sonication temperatures tend to enhance tensile strength, while longer curing durations and shorter sonication periods are consistently linked to improved flexural strength.

- The wide-ranging dataset used in this study makes the models reliable and useful for different types of carbon fiber composites. SHAP beeswarm plots clearly show how each processing parameter affects performance, helping researchers understand the impact of each one. This makes it easier to design high-performing materials while saving time and cost by reducing the need for repeated experiments, especially in the early stages of development.

In summary, by integrating ensemble ML models with interpretable tools like SHAP beeswarm plots, this research not only offers a strong predictive capability but also provides clear insights into how each processing parameter and material property contributes to the final performance. The wide-ranging dataset used in this study ensures the framework's reliability and applicability across various composite designs. This approach enhances predictive accuracy and empowers material scientists to optimize experimental efforts, facilitating the design of high-performance materials. By focusing on the most impactful parameters, this methodology can significantly reduce experimental time and cost, providing a valuable tool for advancing the development of advanced CFRC composite materials.

Funding statement

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Akash Baral: Writing – review & editing, Writing – original draft, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Mostofa Aadeeb Chowdhury:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Md.Jahid Hasan:** Writing – review & editing, Writing – original draft, Validation, Software, Data curation, Conceptualization. **Arda Arefin Nir:** Writing – review & editing, Writing – original draft, Investigation. **Manish Bhadra Arnab:** Writing – review & editing, Writing – original draft, Software, Data curation. **Md. Ashraful Islam:** Writing – review & editing, Writing – original draft, Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors gratefully acknowledge the computational facilities of the Department of Mechanical Engineering, Khulna University of Engineering & Technology, Bangladesh for completing this study.

Data availability

Data will be made available on request.

References

- [1] K.S. Anu, K.A. Vishnumurthy, A. Mahesh, K. Natarajan, Carbon fiber-reinforced, activated carbon-embedded copper oxide nanoparticles/epoxy hybrid composites for EMI shielding in aircraft applications, *Polym. Bull.* 81 (2024) 8723–8750, <https://doi.org/10.1007/s00289-023-05112-w>.
- [2] A. Namdev, A. Telang, R. Purohit, Effect of graphene nano platelets on mechanical and physical properties of carbon fibre/epoxy hybrid composites, *Adv. Mater. Process. Technol.* 8 (2022) 1168–1181, <https://doi.org/10.1080/2374068X.2021.1939557>.
- [3] Y.-C. Chiou, H.-Y. Chou, M.-Y. Shen, Effects of adding graphene nanoplatelets and nanocarbon aerogels to epoxy resins and their carbon fiber composites, *Mater. Des.* 178 (2019) 107869, <https://doi.org/10.1016/j.matdes.2019.107869>.
- [4] F. Liu, S. Deng, J. Zhang, Mechanical properties of epoxy and its carbon fiber composites modified by nanoparticles, *J. Nanomater.* 2017 (2017) 1–9, <https://doi.org/10.1155/2017/8146248>.
- [5] K.P. Srinivasa Perumal, L. Selvarajan, K.P. Manikandan, C. Velmurugan, Mechanical, tribological, and surface morphological studies on the effects of hybrid ilmenite and silicon dioxide fillers on glass fibre reinforced epoxy composites, *J. Mech. Behav. Biomed. Mater.* 146 (2023) 106095, <https://doi.org/10.1016/j.jmbbm.2023.106095>.
- [6] Y. Wang, S.K. Raman Pillai, J. Che, M.B. Chan-Park, High interlaminar shear strength enhancement of carbon fiber/epoxy composite through fiber- and matrix-anchored carbon nanotube networks, *ACS Appl. Mater. Interfaces* 9 (2017) 8960–8966, <https://doi.org/10.1021/acsami.6b13197>.
- [7] D. Sheth, S. Maiti, S. Patel, J. Kandasamy, M.R. Chandan, A. Rahaman, Enhancement of mechanical properties of carbon fiber reinforced epoxy matrix laminated composites with multiwalled carbon nanotubes, *Fuller. Nanotub. Carbon Nanostructures* 29 (2021) 288–294, <https://doi.org/10.1080/1536383X.2020.1839424>.
- [8] G. Spinelli, R. Guarini, T. Bataklijev, L. Guadagno, M. Raimondo, Testing, experimental design, and numerical analysis of nanomechanical properties in epoxy hybrid systems reinforced with carbon nanotubes and graphene nanoparticles, *Polymers* (2024).
- [9] B. Bahtiri, B. Arash, S. Scheffler, M. Jux, R. Rolles, A thermodynamically consistent physics-informed deep learning material model for short fiber/polymer nanocomposites, *Comput. Methods Appl. Mech. Eng.* (2024).
- [10] L. Bhanuprakash, S. Parasuram, S. Varghese, Experimental investigation on graphene oxides coated carbon fibre/epoxy hybrid composites: mechanical and electrical properties, *Compos. Sci. Technol.* 179 (2019) 134–144, <https://doi.org/10.1016/j.compscitech.2019.04.034>.
- [11] N.C. Adak, S. Chhetri, N.H. Kim, N.C. Murmu, P. Samanta, T. Kuila, Static and dynamic mechanical properties of graphene oxide-incorporated woven carbon fiber/epoxy composite, *J. Mater. Eng. Perform.* 27 (2018) 1138–1147, <https://doi.org/10.1007/s11665-018-3201-5>.
- [12] N.D. Alexopoulos, Z. Paragkamian, P. Poulin, S.K. Kourkoulis, Fracture related mechanical properties of low and high graphene reinforcement of epoxy nanocomposites, *Compos. Sci. Technol.* 150 (2017) 194–204, <https://doi.org/10.1016/j.compscitech.2017.07.030>.
- [13] D. Jiang, L. Liu, G. Wu, Q. Zhang, J. Long, Z. Wu, F. Xie, Y. Huang, Mechanical properties of carbon fiber composites modified with graphene oxide in the interphase, *Polym. Compos.* 38 (2017) 2425–2432, <https://doi.org/10.1002/pc.23828>.
- [14] R. Moghimi Monfared, M.R. Ayatollahi, R. Barbaz Isfahani, Synergistic effects of hybrid MWCNT/nanosilica on the tensile and tribological properties of woven carbon fabric epoxy composites, *Theor. Appl. Fract. Mech.* 96 (2018) 272–284, <https://doi.org/10.1016/j.tafmec.2018.05.007>.
- [15] B.H. Sulaiman, A.R. Abdo, A. Erklig, Ö.Y. Bozkurt, M. Bulut, Effectiveness of MULTI-WALLED CARBON NANOTUBE on the improvement of tensile, flexural, and low-velocity impact properties of hybrid K evlar/carbon fiber reinforced epoxy-based composites, *Polym. Compos.* 45 (2024) 7647–7660, <https://doi.org/10.1002/pc.28294>.
- [16] M.R. Zakaria, M.H. Abdul Kudus, H. Md. Akil, M.Z. Mohd Thirumizir, Comparative study of graphene nanoparticle and multiwall carbon nanotube filled epoxy nanocomposites based on mechanical, thermal and dielectric properties, *Compos. B Eng.* 119 (2017) 57–66, <https://doi.org/10.1016/j.compositesb.2017.03.023>.
- [17] M. Alsaadi, M. Bulut, A. Erklig, A. Jabbar, Nano-silica inclusion effects on mechanical and dynamic behavior of fiber reinforced carbon/Kevlar with epoxy resin hybrid composites, *Compos. B Eng.* 152 (2018) 169–179, <https://doi.org/10.1016/j.compositesb.2018.07.015>.
- [18] M.V. De Pours, J. Shariff, K.S. Chakravarthy, K.S. Rao, V.S. Prasad, Y.S. Reddy, G. Kaliyaperumal, R. Venkatesh, G. Saveetha, Influence of silica nanoparticles on mechanical properties of bast fibers made epoxy composite via injection moulding route, *J. Inst. Eng. India Ser. D.* (2024), <https://doi.org/10.1007/s40033-024-00738-0>.
- [19] V. Eskizeybek, H. Ulus, H.B. Kaybal, Ö.S. Şahin, A. Avcı, Static and dynamic mechanical responses of CaCO₃ nanoparticle modified epoxy/carbon fiber nanocomposites, *Compos. B Eng.* 140 (2018) 223–231, <https://doi.org/10.1016/j.compositesb.2017.12.013>.
- [20] K.P. Srinivasa Perumal, L. Selvarajan, P. Mathan Kumar, S. Shriguppikar, Enhancing mechanical and morphological properties of glass fiber reinforced epoxy polymer composites through rutile nanoparticle incorporation, *Prog. Addit. Manuf.* 10 (1) (2025) 831–848, <https://doi.org/10.1007/s40964-024-00675-0>.
- [21] K.P. Srinivasa Perumal, R. Boopathi, P. Saravanan, L. Selvarajan, Effect of zircon and anatase titanium dioxide nanoparticles on glass fibre reinforced epoxy with mechanical and morphological studies, *Ceram. Int.* 49 (13) (2023) 21667–21677, <https://doi.org/10.1016/j.ceramint.2023.03.304>.
- [22] K. Karthik, D. Rajamani, E.P. Venkatesan, M.I. Shajahan, A.A. Rajhi, A. Aabid, M. Baig, B. Saleh, Experimental investigation of the mechanical properties of carbon/basalt/sic nanoparticle/polyester hybrid composite materials, *Crystals* 13 (2023) 415, <https://doi.org/10.3390/cryst13030415>.
- [23] G. Barra, L. Guadagno, L. Vertuccio, B. Simonet, B. Santos, M. Zarrelli, M. Arena, M. Viscardi, Different methods of dispersing carbon nanotubes in epoxy resin and initial evaluation of the obtained nanocomposite as a matrix of carbon fiber reinforced laminate in terms of vibroacoustic performance and flammability, *Materials* (2019).
- [24] Mrázová, M.: Advanced composite materials of the future in aerospace industry. (2013).
- [25] Chwal, M., Muc, A.: FEM micromechanical modeling of nanocomposites with carbon nanotubes. (2021).
- [26] N. Hameed, N. Salim, T. Hanley, M. Sona, B. Fox, Q. Guo, Individual dispersion of carbon nanotubes in epoxy via a novel dispersion-curing approach using ionic liquids, *Phys. Chem. Chem. Phys. - PCCP.* (2013).
- [27] S.M. Shahbaz, P. Mehrotra, H. Kalita, S. Sharma, N. Naik, D.J. Noronha, N. Shetty, Effect of Al₂O₃ and SiC nano-fillers on the mechanical properties of carbon fiber-reinforced epoxy hybrid composites, *J. Compos. Sci.* 7 (2023) 133, <https://doi.org/10.3390/jcs7040133>.
- [28] Ünal, H.Y., Öner, G., Pekbey, Y.: Comparison of the Experimental Mechanical Properties and DMA Measurement of Nanoclay Hybrid Composites. (2017).
- [29] M. Bhattacharya, Polymer nanocomposites—a comparison between carbon nanotubes, graphene, and clay as nanofillers, *Materials* (2016).
- [30] B. Arash, W. Exner, R. Rolles, Effect of moisture on the nonlinear viscoelastic fracture behavior of polymer nanocomposites: a finite deformation phase-field model, *Eng. Comput.* (2022).
- [31] S. Tamayo-Vegas, A. Muhsan, C. Liu, M. Tarfaoui, K. Lafdi, The effect of agglomeration on the electrical and mechanical properties of polymer matrix nanocomposites reinforced with carbon nanotubes, *Polymers* (2022).
- [32] M. Chwal, A. Muc, Design of reinforcement in nano- and microcomposites, *Materials* (2019).
- [33] Tserpes, K.: Strength Prediction of Composite Materials from Nano- to Macro-scale. (2011).
- [34] H.A. Mahmud, M. Radue, S. Chinkanjanarot, G. Odegard, Multiscale modeling of epoxy-based nanocomposites reinforced with functionalized and non-functionalized graphene nanoplatelets, *Polymers* (2021).
- [35] A. Gbaguidi, S. Namilaie, D. Kim, Synergy effect in hybrid nanocomposites based on carbon nanotubes and graphene nanoplatelets, *Nanotechnology* (2020).
- [36] Al-Kharusi, M., Alzebeid, K., Pervez, T.: An Atomistic-Based Continuum Modeling for Evaluation of Effective Elastic Properties of Single-Walled Carbon Nanotubes. (2016).

- [37] Karevan, M., Pucha, R., Bhuiyan, A., Kalaitzidou, K.: Effect of Interphase Modulus and Nanofiller Agglomeration on the Tensile Modulus of Graphite Nanoplatelets and Carbon Nanotube Reinforced Polypropylene Nanocomposites. (2010).
- [38] Lee, J., Park, S., Kim, Y.-H.: Improvement of Interfacial Adhesion of Incorporated Halloysite-Nanotubes in Fiber-Reinforced Epoxy-Based Composites. (2017).
- [39] Z. Liu, C.T. Wu, Exploring the 3D architectures of deep material network in data-driven multiscale mechanics, *J. Mech. Phys Solids* (2019).
- [40] E. Demir, M. McDermott, C. Kim Il, C. Ayranci, Towards better understanding the stiffness of nanocomposites via parametric study of an analytical model modeling parameters and experiments, *J. Compos. Mater.* (2023).
- [41] Yu, H., Qin, Z., Yan, C., Yao, J.: Green Nanocomposites Based on Functionalized Cellulose Nanocrystals: A Study on the Relationship between Interfacial Interaction and Property Enhancement. (2014).
- [42] Y. He, J. Zhu, W. Wang, H. Ni, Surface modification of cellulose nanocrystals with different acid anhydrides for improved dispersion in poly(butylene succinate), *RSC Adv.* (2018).
- [43] Giannelis, E.: Polymer Layered Silicate Nanocomposites. (1996).
- [44] C.L. Grant, H. Hadavinia, N.A. Williams, S. Koutsonas, Calculation of interphase elastic properties of esterified cellulose nanocrystal/epoxy bio-nanocomposites using analytical and finite element methods, *Sci. Prog. (New Haven)*. (2025).
- [45] Ji, X., Jing, J., Jiang, W., Jiang, B.: Tensile modulus of polymer nanocomposites. (2002).
- [46] V. Gavande, S. Nagappan, W.-K. Lee, Considering electrospun nanofibers as a filler network in electrospun nanofiber-reinforced composites to predict the tensile strength and young's modulus of nanocomposites: a modeling study, *Polymers* (2022).
- [47] Sharma, K., Shukla, M.: Three-phase carbon fiber amine functionalized carbon nanotubes epoxy composite: processing, characterisation, and multiscale modeling. (2014).
- [48] S. Uthale, N.A. Dhamal, D.K. Shinde, A. Kelkar, Polymeric hybrid nanocomposites processing and finite element modeling: an overview, *Sci. Prog. (New Haven)*. (2021).
- [49] M. Javadinejad, M. Mashayekhi, M. Karevan, H. Hadavinia, Using the equivalent fiber approach in two-scale modeling of the elastic behavior of carbon nanotube/epoxy nanocomposite, *Nanomaterials* (2018).
- [50] Shah, M.A., Yunus, M., Rani, M., Mirza, W., Sani, M.: An Improved Method for Dynamic Behaviour Prediction of Carbon Fibre Reinforced Epoxy (CFRE) using Finite Element Model Updating. (2021).
- [51] S. Yang, Z. Cui, J. Qu, A coarse-grained model for epoxy molding compound, *J. Phys. Chem. B* (2014).
- [52] S. Ferdousi, Q. Chen, M. Soltani, J. Zhu, P. Cao, W. Choi, R. Advíncula, Y. Jiang, Characterize traction-separation relation and interfacial imperfections by data-driven machine learning models, *Sci. Rep.* (2021).
- [53] S.S. Sorour, C.A. Saleh, M. Shazly, A review on machine learning implementation for predicting and optimizing the mechanical behaviour of laminated fiber-reinforced polymer composites, *Heliyon* (2024).
- [54] E. Koumoulos, G. Konstantopoulos, C. Charitidis, Applying machine learning to nanoindentation data of (nano-) enhanced, *Composites* (2019).
- [55] A. Challapalli, G. Li, 3D printable biomimetic rod with superior buckling resistance designed by machine learning, *Sci Rep.* (2020).
- [56] S. Pruksawan, G. Lambard, S. Samitsu, K. Sodeyama, M. Naito, Prediction and optimization of epoxy adhesive strength from a small dataset through active learning, *Sci. Technol. Adv. Mater.* (2019).
- [57] B.S. S., Hamaekers, J.: Feature-based prediction of properties of cross-linked epoxy polymers by molecular dynamics and machine learning techniques. (2023).
- [58] Y. Zhao, Z. Chen, X. Jian, A high-generalizability machine learning framework for analyzing the homogenized properties of short fiber-reinforced polymer composites, *Polymers* (2023).
- [59] I.P. Malashin, V. Tynchenko, A.P. Gantimurov, V. Nelyub, A.S. Borodulin, Boosting-based machine learning applications in polymer science: a review, *Polymers* (2025).
- [60] J. Zhao, Z. Guo, Q. Lyu, B. Wang, Prediction of residual compressive strength after impact based on acoustic emission characteristic parameters, *Polymers* (2024).
- [61] M. Rohatgi, A. Kordijazi, Application of machine learning to mechanical properties of copper-graphene composites, *MRS Commun.* (2022).
- [62] S.-J. Huang, Y. Adityawardhana, J. Sanjaya, Predicting mechanical properties of magnesium matrix composites with regression models by machine learning, *J. Compos. Sci.* (2023).
- [63] S. Che, G. Qu, G. Wang, Y. Hao, J. Sun, J. Ding, A review of the biomimetic structural design of sandwich composite materials, *Polymers* (2024).
- [64] M. Yuan, H. Zhao, S. Liu, H. Ren, B. Zhang, X. Sun, J. Chen, Determination of cohesive parameters for fibre-reinforced composite interfaces based on finite element analysis and machine learning, *J. Compos. Mater.* (2022).
- [65] A. Sharma, T. Mukhopadhyay, S. Rangappa, S. Siengchin, V. Kushvaha, Advances in computational intelligence of polymer composite materials: machine learning assisted modeling, *Analysis and Design. Arch. Comput. Methods Eng.* (2021).
- [66] J.B. Deb, J. Gou, H. Song, C. Maiti, Machine learning approaches for predicting the ablation performance of ceramic matrix composites, *J. Compos. Sci.* (2024).
- [67] G. Pilania, C. Wang, X. Jiang, S. Rajasekaran, R. Ramprasad, Accelerating materials property predictions using machine learning, *Sci. Rep.* (2013).
- [68] R. Karamov, I. Akhatov, I. Sergeichev, Prediction of fracture toughness of pultruded composites based on supervised machine learning, *Polymers* (2022).
- [69] S.A. Shaikh, M. Taufique, B. Kranthi, S.S. Kulkarni, F. Hale, J. Oleson, R. Devanathan, A. Soulam, Finite element analysis and machine learning guided design of carbon fiber organosheet-based battery enclosures for crashworthiness, *arXiv.org* (2023).
- [70] Z. Khan, A. Gul, A. Perperoglou, M. Miftahuddin, O. Mahmoud, W. Adler, B. Lausen, Ensemble of optimal trees, random forest and random projection ensemble classification, *Adv. Data Anal. Classif.* (2019).
- [71] J. Wang, H. Kwon, H.S. Kim, B. Lee, A neural network model for high entropy alloy design, *Npj Comput. Mater.* (2023).
- [72] X. Liu, T. Liu, P. Feng, Long-term performance prediction framework based on XGBoost decision tree for pultruded FRP composites exposed to water, humidity and alkaline solution, *Compos. Struct.* 284 (2022) 115184, <https://doi.org/10.1016/j.compstruct.2022.115184>.
- [73] M.N. Uddin, Al-Amin, S. Hossain, Revolutionizing engineered cementitious composite materials (ECC): the impact of XGBoost-SHAP analysis on polyvinyl alcohol (PVA) based ECC predictions, *Low-Carbon Mater. Green Constr.* 2 (2024) 11, <https://doi.org/10.1007/s44242-024-00042-w>.
- [74] A.J. Mohammed, A.S. Mohammed, A.S. Mohammed, Prediction of tribological properties of UHMWPE/SiC polymer composites using machine learning techniques, *Polymers* (2023).
- [75] N. Amor, M. Noman, M. Petrú, Classification of textile polymer composites: recent trends and challenges, *Polymers* (2021).
- [76] E. Champa-Bujaico, A. Díez-Pascual, P. García-Díaz, Synthesis and characterization of polyhydroxyalkanoate/graphene oxide/nanoclay bionanocomposites: experimental results and theoretical predictions via machine learning models, *Biomolecules* (2023).
- [77] J. León-Becerra, O. González-Estrada, H. Sánchez-Acevedo, Comparison of models to predict mechanical properties of FR-AM composites and a fractographical study, *Polymers* (2022).
- [78] Y. Yilmaz, T. Çakmak, Z. Kurt, I. Ustaş, Predicting mechanical properties in geopolymers mortars, including novel precursor combinations, *Through XGBoost Method. Arab. J. Sci. Eng.* (2024).
- [79] D.K. Lim, K.B. Mustapha, C.P. Pagwiwoko, Delamination detection in composite plates using random forests, *Compos. Struct.* 278 (2021) 114676, <https://doi.org/10.1016/j.compstruct.2021.114676>.
- [80] T. Adithiyaa, D. Chandramohan, T. Sathish, Optimal prediction of process parameters by GWO-KNN in stirring-squeeze casting of AA2219 reinforced metal matrix composites, *Mater. Today Proc.* 21 (2020) 1000–1007, <https://doi.org/10.1016/j.matpr.2019.10.051>.
- [81] B. Rajiv, P. Kalos, P. Pantawane, V. Chougule, V. Chavan, Classification of damages in composite material using multi-support vector machine, *J. Inst. Eng. India Ser. C*. 103 (2022) 655–661, <https://doi.org/10.1007/s40032-022-00811-1>.
- [82] V. Daghighi, S. Bakhtiari Ramezani, H. Daghighi, T.E. Lacy Jr., Explainable artificial intelligence prediction of defect characterization in composite materials, *Compos. Sci. Technol.* 256 (2024) 110759, <https://doi.org/10.1016/j.compscitech.2024.110759>.
- [83] M. Yossef, M. Noureldin, A. Alqabbany, Explainable artificial intelligence framework for FRP composites design, *Compos. Struct.* 341 (2024) 118190, <https://doi.org/10.1016/j.compstruct.2024.118190>.
- [84] M. Hematiabhar, M. Kharun, A.N. Beskoplyny, S.A. Stel'makh, E.M. Shcherban', I. Razveeva, Analysis of Models to Predict Mechanical Properties of High-Performance and Ultra-High-Performance Concrete using Machine Learning, *J. Compos. Sci.* 8 (2024) 287, <https://doi.org/10.3390/jcs8080287>.
- [85] X. Zheng, Y. Xie, X. Yang, M.N. Amin, S. Nazar, S.A. Khan, F. Althoei, A. F. Deifalla, A data-driven approach to predict the compressive strength of alkali-activated materials and correlation of influencing parameters using SHapley Additive exPlanations (SHAP) analysis, *J. Mater. Res. Technol.* 25 (2023) 4074–4093, <https://doi.org/10.1016/j.jmrt.2023.06.207>.
- [86] P.B. Pande, S.W. Dhengare, J.M. Raut, R.M. Bhagat, B.V. Bahoria, N. Shelke, S. Nair, V.S. Vairagade, Integrated hybrid machine learning techniques and multiscale modeling towards evaluating the influence of nanomaterial on strength of concrete, *Multiscale Multidiscip. Model. Exp. Des.* 8 (2025) 26, <https://doi.org/10.1007/s41939-024-00588-z>.
- [87] "sklearn.model.selection.GroupKFold — sklearn 1.0.2 documentation." Accessed: Aug. 05, 2025. [Online]. Available: <https://scikit-learn.org/1.0/modules/generated/sklearn.model.selection.GroupKFold.html>.
- [88] L. Breiman, Random forests, *Mach. Learn.* 45 (1) (Oct. 2001) 5–32, <https://doi.org/10.1023/A:1010933404324>.
- [89] T. Chen, C. Guestrin, XGBoost: a scalable tree boosting, *System* (2016), <https://doi.org/10.48550/ARXIV.1603.02754>.
- [90] J.H. Friedman, <http://www.jstor.org> greedy function approximation: a Gradient Boosting Machine, *Ann. Stat.* 29 (2001) 1189–1232.
- [91] G. James, D. Witten, T. Hastie, R. Tibshirani, *An Introduction to Statistical Learning: with Applications in R*, Springer, US, New York, NY, 2021.
- [92] X. Chen, H. Xu, D. Liu, C. Yan, Y. Zhu, A novel and facile fabrication of polyphosphazene nanotube/carbon fiber multiscale hybrid reinforcement and its enhancing effect on the interfacial properties of epoxy composites, *Compos. Sci. Technol.* 169 (2019) 34–44, <https://doi.org/10.1016/j.compscitech.2018.10.034>.
- [93] C. Kurtulus, M. Kuyumcu, M. Ciftci, M.A. Tasdelen, Influence of POSS nanoparticles on the microstructure and mechanical properties of carbon fiber reinforced epoxy hybrid composites, *Polym. Compos.* 42 (2021) 4056–4064, <https://doi.org/10.1002/pc.26116>.
- [94] H. Kaftelen-Odabaşı, A. Odabaşı, M. Özdemir, M. Baydoğan, A study on graphene reinforced carbon fiber epoxy composites: investigation of electrical, flexural, and dynamic mechanical properties, *Polym. Compos.* 44 (2023) 121–135, <https://doi.org/10.1002/pc.27031>.

- [95] H. Chamkouri, M. Pooremaei, H. Namazi, Carbon fiber/epoxy resin/ α -aluminum oxide nanocomposites; fabrication, mechanical and thermal analysis, Iran. Polym. J. 30 (2021) 523–533, <https://doi.org/10.1007/s13726-021-00909-w>.
- [96] X. Yao, X. Gao, J. Jiang, C. Xu, C. Deng, J. Wang, Comparison of carbon nanotubes and graphene oxide coated carbon fiber for improving the interfacial properties of carbon fiber/epoxy composites, Compos. B Eng. 132 (2018) 170–177, <https://doi.org/10.1016/j.compositesb.2017.09.012>.
- [97] R.L. Zhang, B. Gao, Q.H. Ma, J. Zhang, H.Z. Cui, L. Liu, Directly grafting graphene oxide onto carbon fiber and the effect on the mechanical properties of carbon fiber composites, Mater. Des. 93 (2016) 364–369, <https://doi.org/10.1016/j.matdes.2016.01.003>.
- [98] S.M. Shahabaz, P.K. Shetty, N. Shetty, S. Sharma, S. Divakara Shetty, N. Naik, Effect of alumina and silicon carbide nanoparticle-infused polymer matrix on mechanical properties of unidirectional carbon fiber-reinforced polymer, J. Compos. Sci. 6 (2022) 381, <https://doi.org/10.3390/jcs6120381>.
- [99] K. Mishra, K.P. Bastola, R.P. Singh, R. Vaidyanathan, Effect of graphene oxide on the interlaminar fracture toughness of carbon fiber/epoxy composites, Polym. Eng. Sci. 59 (2019) 1199–1208, <https://doi.org/10.1002/pen.25100>.
- [100] E. Kosedag, R. Ekici, N. Yildiz, U. Caliskan, Effect of SiC and graphene nanoparticles on the mechanical properties of carbon fiber-reinforced epoxy composites, Polym. Compos. 44 (2023) 8578–8588, <https://doi.org/10.1002/pc.27720>.
- [101] M. Ayatollahi, R. Barbaz Isfahani, R. Moghimi Monfared, Effects of multi-walled carbon nanotube and nanosilica on tensile properties of woven carbon fabric-reinforced epoxy composites fabricated using VARIM, J. Compos. Mater. 51 (2017) 4177–4188, <https://doi.org/10.1177/0021998317699982>.
- [102] Ş. Yazman, M. Uyaner, F. Karabörk, A. Akdemir, Effects of nano reinforcing/matrix interaction on chemical, thermal and mechanical properties of epoxy nanocomposites, J. Compos. Mater. 55 (2021) 4257–4272, <https://doi.org/10.1177/00219983211037059>.
- [103] S.A. Mirsalehi, A.A. Youzbashi, A. Sazgar, Enhancement of out-of-plane mechanical properties of carbon fiber reinforced epoxy resin composite by incorporating the multi-walled carbon nanotubes, SN Appl. Sci. 3 (2021) 630, <https://doi.org/10.1007/s42452-021-04624-2>.
- [104] S.Z. Haeri, M. Asghari, B. Ramezanzadeh, Enhancement of the mechanical properties of an epoxy composite through inclusion of graphene oxide nanosheets functionalized with silica nanoparticles through one and two steps sol-gel routes, Prog. Org. Coat. 111 (2017) 1–12, <https://doi.org/10.1016/j.porgcoat.2017.05.003>.
- [105] A. Namdev, A. Telang, R. Purohit, Experimental investigation on mechanical and wear properties of GNP/Carbon fiber/epoxy hybrid composites, Mater. Res. Express 9 (2022) 025303, <https://doi.org/10.1088/2053-1591/ac4e3f>.
- [106] S. Mandal, K. Samanta, K. Manna, S. Kumar, S. Bose, GO-tagged PEI sizing agent imparts self-healing and excellent mechanical properties to carbon fiber reinforced epoxy laminates, Nanoscale 16 (2024) 6984–6998, <https://doi.org/10.1039/D3NR06047K>.
- [107] Y. Li, H. Zhang, Z. Huang, E. Bilotti, T. Peijs, Graphite nanoplatelet modified epoxy resin for carbon fibre reinforced plastics with enhanced properties, J. Nanomater. 2017 (2017) 1–10, <https://doi.org/10.1155/2017/5194872>.
- [108] B. Song, T. Wang, H. Sun, H. Liu, X. Mai, X. Wang, L. Wang, N. Wang, Y. Huang, Z. Guo, Graphitic carbon nitride (g-C₃N₄) interfacially strengthened carbon fiber epoxy composites, Compos. Sci. Technol. 167 (2018) 515–521, <https://doi.org/10.1016/j.compscitech.2018.08.031>.
- [109] N.I. Khan, S. Halder, S. Das, M.S. Goyat, Graphitic nanoparticles functionalized with epoxy moiety for enhancing the mechanical performance of hybrid carbon fiber reinforced polymer laminated composites, Polym. Compos. 42 (2021) 678–692, <https://doi.org/10.1002/pc.25857>.
- [110] Y. Wu, Y. Zhang, Z. Yang, X. Huang, S. Chen, J. Li, G. Huang, W. Wei, G. Gao, G. Wu, In situ growth of silver nanoparticles on carbon fiber by plasma-liquid interaction to improve performance of epoxy composites, J. Appl. Phys. 136 (2024) 065304, <https://doi.org/10.1063/5.0218525>.
- [111] S. Shahbakhsh, E. Tohidlou, H. Khosravi, Influence of modified carbonate calcium nanoparticles on the mechanical properties of carbon fiber/epoxy composites, J. Text. Inst. 111 (2020) 550–554, <https://doi.org/10.1080/00405000.2019.1651155>.
- [112] S. Singh, V.K. Srivastava, R. Prakash, Influences of carbon nanofillers on mechanical performance of epoxy resin polymer, Appl. Nanosci. 5 (2015) 305–313, <https://doi.org/10.1007/s13204-014-0319-0>.
- [113] F.H. Chowdhury, M.V. Hosur, S. Jeelani, Investigations on the thermal and flexural properties of plain weave carbon/epoxy-nanoclay composites by hand-layup technique, J. Mater. Sci. 42 (2007) 2690–2700, <https://doi.org/10.1007/s10853-006-1370-3>.
- [114] P.P. Prasanthi, M.S.R.N. Kumar, M.S. Chowdary, V.V.V. Madhav, K.K. Saxena, K. A. Mohammed, M.I. Khan, G. Upadhyay, S.M. Eldin, Mechanical properties of carbon fiber reinforced with carbon nanotubes and graphene filled epoxy composites: experimental and numerical investigations, Mater. Res. Express 10 (2023) 025308, <https://doi.org/10.1088/2053-1591/acaef5>.
- [115] W. Lin, Q.Q. Shi, H. Chen, J.N. Wang, Mechanical properties of carbon nanotube fibers reinforced epoxy resin composite films prepared by wet winding, Carbon 153 (2019) 308–314, <https://doi.org/10.1016/j.carbon.2019.07.002>.
- [116] M.V. Burkov, A.V. Eremin, Mechanical properties of carbon-fiber-reinforced EPOXY COMPOSITES modified by carbon micro- and NANOFILLERS, Polym. Compos. 42 (2021) 4265–4276, <https://doi.org/10.1002/pc.26144>.
- [117] Z. Zhao, K. Teng, N. Li, X. Li, Z. Xu, L. Chen, J. Niu, H. Fu, L. Zhao, Y. Liu, Mechanical, thermal and interfacial performances of carbon fiber reinforced composites flavored by carbon nanotube in matrix/interface, Compos. Struct. 159 (2017) 761–772, <https://doi.org/10.1016/j.compstruct.2016.10.022>.
- [118] W. Chen, Y. Liu, Z. Jiang, L. Tang, Z. Liu, L. Zhou, Modeling of compressive strength for unidirectional fiber reinforced composites with nanoparticle modified epoxy matrix, Materials 12 (2019) 3897, <https://doi.org/10.3390/ma12233897>.
- [119] W. Qin, F. Vautard, L.T. Drzal, J. Yu, Modifying the carbon fiber-epoxy matrix interphase with graphite nanoplatelets, Polym. Compos. 37 (2016) 1549–1556, <https://doi.org/10.1002/pc.23325>.
- [120] C. Li, H. Guo, X. Tian, T. He, Size-dependent thermo-electromechanical responses analysis of multi-layered piezoelectric nanoplates for vibration control, Compos. Struct. 225 (2019) 111112, <https://doi.org/10.1016/j.compstruct.2019.111112>.
- [121] Y. Huang, I.A. Kinloch, C. Vallés, Static and dynamic mechanical behavior of carbon fiber reinforced plastic (CFRP) single-lap shear joints joule-bonded with conductive epoxy nanocomposites, J. Compos. Sci. 8 (2024) 112, <https://doi.org/10.3390/jcs8030112>.
- [122] “automeris.io: Computer vision assisted data extraction from charts using WebPlotDigitizer.” Accessed: Aug. 05, 2025. [Online]. Available: <https://automeris.io/>.
- [123] “LabelEncoder — scikit-learn 1.7.1 documentation.” Accessed: Aug. 05, 2025. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.LabelEncoder.html>.
- [124] “StandardScaler — scikit-learn 1.7.1 documentation.” Accessed: Aug. 05, 2025. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html>.